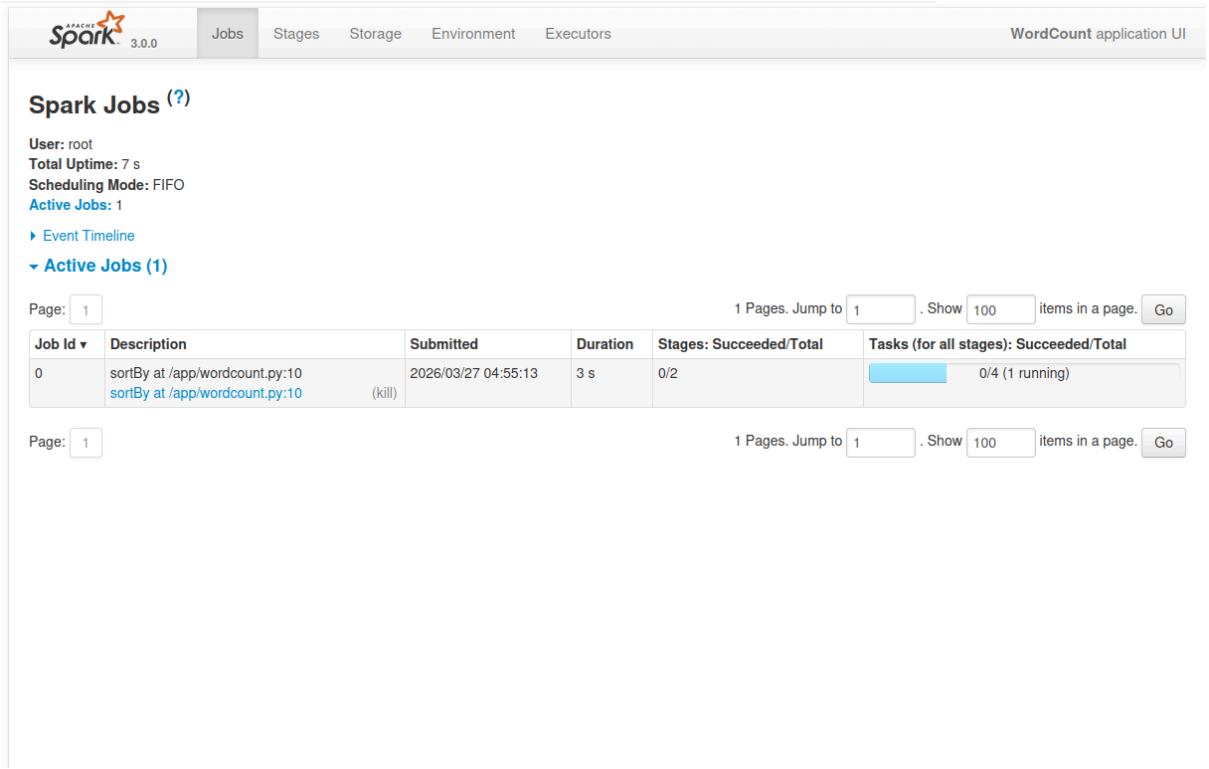
Bài 1:

Thực hiện bài toán WordCount sử dụng RDD Spark trên cụm 3 máy

Chụp ảnh kết quả + các trình Spark sinh ra

```
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Nam-3/BigData/spark-cluster$ docker exec spark-master /spark/bin/spark-submit \
  --master spark://spark-master:7077 \
  --num-executors 3 \
  --executor-cores 1 \
  --executor-memory 512m \ …
26/03/27 04:55:18 INFO DAGScheduler: Registering RDD 9 (sortBy at /app/wordcount.py:10) as input to shuffle 1
26/03/27 04:55:18 INFO DAGScheduler: Got job 2 (collect at /app/wordcount.py:13) with 2 output partitions
26/03/27 04:55:18 INFO DAGScheduler: Final stage: ResultStage 6 (collect at /app/wordcount.py:13)
26/03/27 04:55:18 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 5)
26/03/27 04:55:18 INFO DAGScheduler: Missing parents: List(ShuffleMapStage 5)
26/03/27 04:55:18 INFO DAGScheduler: Submitting ShuffleMapStage 5 (PairwiseRDD[9] at sortBy at /app/wordcount.py:10), which has no missing parents
26/03/27 04:55:18 INFO MemoryStore: Block broadcast_4 stored as values in memory (estimated size 11.4 KiB, free 365.9 MiB)
26/03/27 04:55:18 INFO MemoryStore: Block broadcast_4_piece0 stored as bytes in memory (estimated size 7.1 KiB, free 365.9 MiB)
26/03/27 04:55:18 INFO BlockManagerInfo: Added broadcast_4_piece0 in memory on 33507b646d2d:37103 (size: 7.1 KiB, free: 366.2 MiB)
26/03/27 04:55:18 INFO SparkContext: Created broadcast 4 from broadcast at DAGScheduler.scala:1200
26/03/27 04:55:18 INFO DAGScheduler: Submitting 2 missing tasks from ShuffleMapStage 5 (PairwiseRDD[9] at sortBy at /app/wordcount.py:10) (first 15 tasks are for partitions Ve
))
26/03/27 04:55:18 INFO TaskSchedulerImpl: Adding task set 5.0 with 2 tasks
26/03/27 04:55:18 INFO TaskSetManager: Starting task 0.0 in stage 5.0 (TID 6, 172.20.0.7, executor 1, partition 0, NODE_LOCAL, 7136 bytes)
26/03/27 04:55:18 INFO TaskSetManager: Starting task 1.0 in stage 5.0 (TID 7, 172.20.0.6, executor 2, partition 1, NODE_LOCAL, 7136 bytes)
26/03/27 04:55:18 INFO BlockManagerInfo: Added broadcast_4_piece0 in memory on 172.20.0.7:37669 (size: 7.1 KiB, free: 93.2 MiB)
26/03/27 04:55:18 INFO BlockManagerInfo: Added broadcast_4_piece0 in memory on 172.20.0.6:39223 (size: 7.1 KiB, free: 93.2 MiB)
26/03/27 04:55:18 INFO TaskSetManager: Finished task 0.0 in stage 5.0 (TID 6) in 158 ms on 172.20.0.7 (executor 1) (1/2)
26/03/27 04:55:18 INFO TaskSetManager: Finished task 1.0 in stage 5.0 (TID 7) in 174 ms on 172.20.0.6 (executor 2) (2/2)
26/03/27 04:55:18 INFO TaskSchedulerImpl: Removed TaskSet 5.0, whose tasks have all completed, from pool
26/03/27 04:55:18 INFO DAGScheduler: ShuffleMapStage 5 (sortBy at /app/wordcount.py:10) finished in 0.191 s
26/03/27 04:55:18 INFO DAGScheduler: looking for newly runnable stages
26/03/27 04:55:18 INFO DAGScheduler: running: Set()
26/03/27 04:55:18 INFO DAGScheduler: waiting: Set(ResultStage 6)
26/03/27 04:55:18 INFO DAGScheduler: failed: Set()
26/03/27 04:55:18 INFO DAGScheduler: Submitting ResultStage 6 (PythonRDD[12] at collect at /app/wordcount.py:13), which has no missing parents
26/03/27 04:55:18 INFO MemoryStore: Block broadcast_5 stored as values in memory (estimated size 8.3 KiB, free 365.9 MiB)
26/03/27 04:55:18 INFO MemoryStore: Block broadcast_5_piece0 stored as bytes in memory (estimated size 4.9 KiB, free 365.9 MiB)
26/03/27 04:55:18 INFO BlockManagerInfo: Added broadcast_5_piece0 in memory on 33507b646d2d:37103 (size: 4.9 KiB, free: 366.2 MiB)
26/03/27 04:55:18 INFO SparkContext: Created broadcast 5 from broadcast at DAGScheduler.scala:1200
26/03/27 04:55:18 INFO DAGScheduler: Submitting 2 missing tasks from ResultStage 6 (PythonRDD[12] at collect at /app/wordcount.py:13) (first 15 tasks are for partitions Vector
26/03/27 04:55:18 INFO TaskSchedulerImpl: Adding task set 6.0 with 2 tasks
26/03/27 04:55:18 INFO TaskSetManager: Starting task 0.0 in stage 6.0 (TID 8, 172.20.0.7, executor 1, partition 0, NODE_LOCAL, 7147 bytes)
26/03/27 04:55:18 INFO TaskSetManager: Starting task 1.0 in stage 6.0 (TID 9, 172.20.0.6, executor 2, partition 1, NODE_LOCAL, 7147 bytes)
26/03/27 04:55:18 INFO BlockManagerInfo: Added broadcast_5_piece0 in memory on 172.20.0.6:39223 (size: 4.9 KiB, free: 93.2 MiB)
26/03/27 04:55:18 INFO BlockManagerInfo: Added broadcast_5_piece0 in memory on 172.20.0.7:37669 (size: 4.9 KiB, free: 93.2 MiB)
26/03/27 04:55:18 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 1 to 172.20.0.6:49606
26/03/27 04:55:18 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 1 to 172.20.0.7:56816
26/03/27 04:55:18 INFO TaskSetManager: Finished task 1.0 in stage 6.0 (TID 9) in 117 ms on 172.20.0.6 (executor 2) (1/2)
26/03/27 04:55:19 INFO TaskSetManager: Finished task 0.0 in stage 6.0 (TID 8) in 153 ms on 172.20.0.7 (executor 1) (2/2)
26/03/27 04:55:19 INFO TaskSchedulerImpl: Removed TaskSet 6.0, whose tasks have all completed, from pool
26/03/27 04:55:19 INFO DAGScheduler: ResultStage 6 (collect at /app/wordcount.py:13) finished in 0.185 s
26/03/27 04:55:19 INFO DAGScheduler: Job 2 is finished. Cancelling potential speculative or zombie tasks for this job
26/03/27 04:55:19 INFO TaskSchedulerImpl: Killing all running tasks in stage 6: Stage finished
```

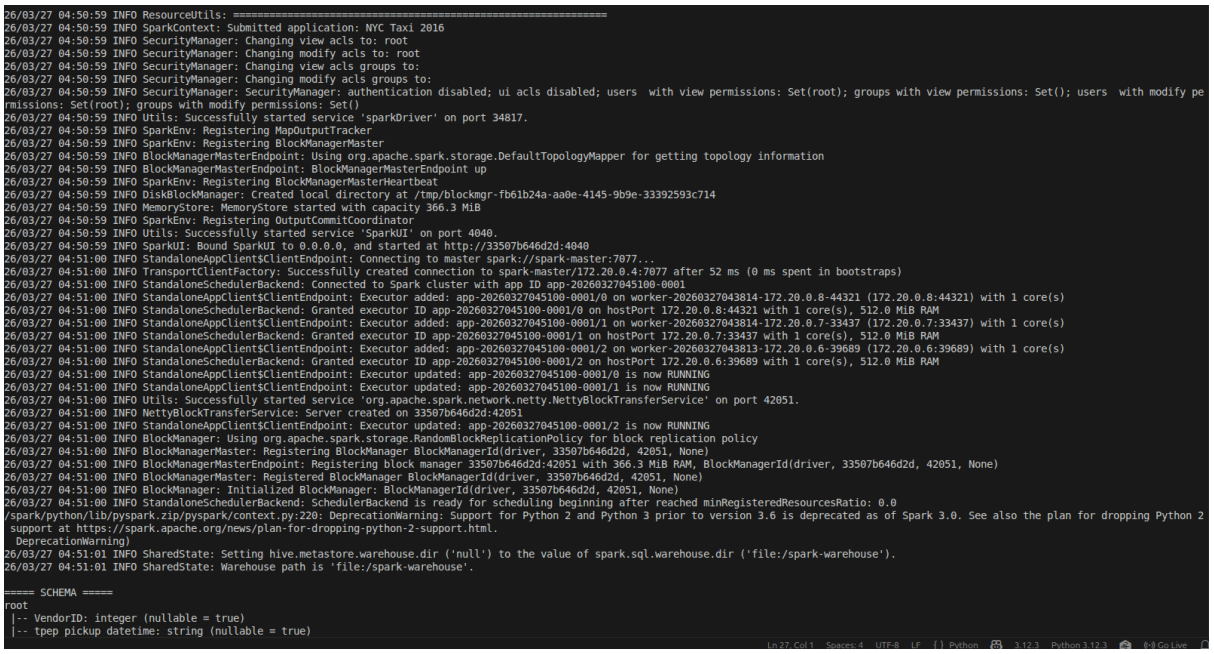
```
vanvan@vanvan-Latitude-9520:/media/vanvan/D/Study in UET/Nam-3/BigData/spark-cluster$ docker exec spark-master /spark/bin/spark-submit \
  --master spark://spark-master:7077 \
  --num-executors 3 \
  --executor-cores 1 \
  --executor-memory 512m \ …
multiple : 1
speed : 1
large : 1
use : 1
java : 1
stream : 1
open : 1
fast : 1
run : 1
widely : 1
resilient : 1
nodes : 1
supports : 1
into : 1
columns : 1
real : 1
handle : 1
python : 1
collection : 1
uses : 1
building : 1
querying : 1
industry : 1
allows : 1
daily : 1
algorithms : 1
popular : 1
graphx : 1
requires : 1
etl : 1
==========================
26/03/27 04:55:19 INFO SparkUI: Stopped Spark web UI at http://33507b646d2d:4040
26/03/27 04:55:19 INFO StandaloneSchedulerBackend: Shutting down all executors
26/03/27 04:55:19 INFO CoarseGrainedSchedulerBackend$DriverEndpoint: Asking each executor to shut down
26/03/27 04:55:19 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/03/27 04:55:19 INFO MemoryStore: MemoryStore cleared
26/03/27 04:55:19 INFO BlockManager: BlockManager stopped
26/03/27 04:55:19 INFO BlockManagerMaster: BlockManagerMaster stopped
26/03/27 04:55:19 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/03/27 04:55:19 INFO SparkContext: Successfully stopped SparkContext
26/03/27 04:55:20 INFO ShutdownHookManager: Shutdown hook called
26/03/27 04:55:20 INFO ShutdownHookManager: Deleting directory /tmp/spark-b46dfd09-3a65-4581-8ba3-a9f5bbb052d1
26/03/27 04:55:20 INFO ShutdownHookManager: Deleting directory /tmp/spark-b46dfd09-3a65-4581-8ba3-a9f5bbb052d1/pyspark-7ff3a1e0-962f-40c0-baa7-0e8823fea668
26/03/27 04:55:20 INFO ShutdownHookManager: Deleting directory /tmp/spark-250b76f0-55e3-4dea-9e0b-a3d240210f10
```



Bài 2: Tiến hành sử dụng DataFrame trên dữ liệu [NYC Taxi Data CSV Links to an external site.](#).

Ví dụ: Tính trung bình số tiền tip theo từng loại hình thanh toán

Chụp ảnh kết quả + các trình Spark sinh ra



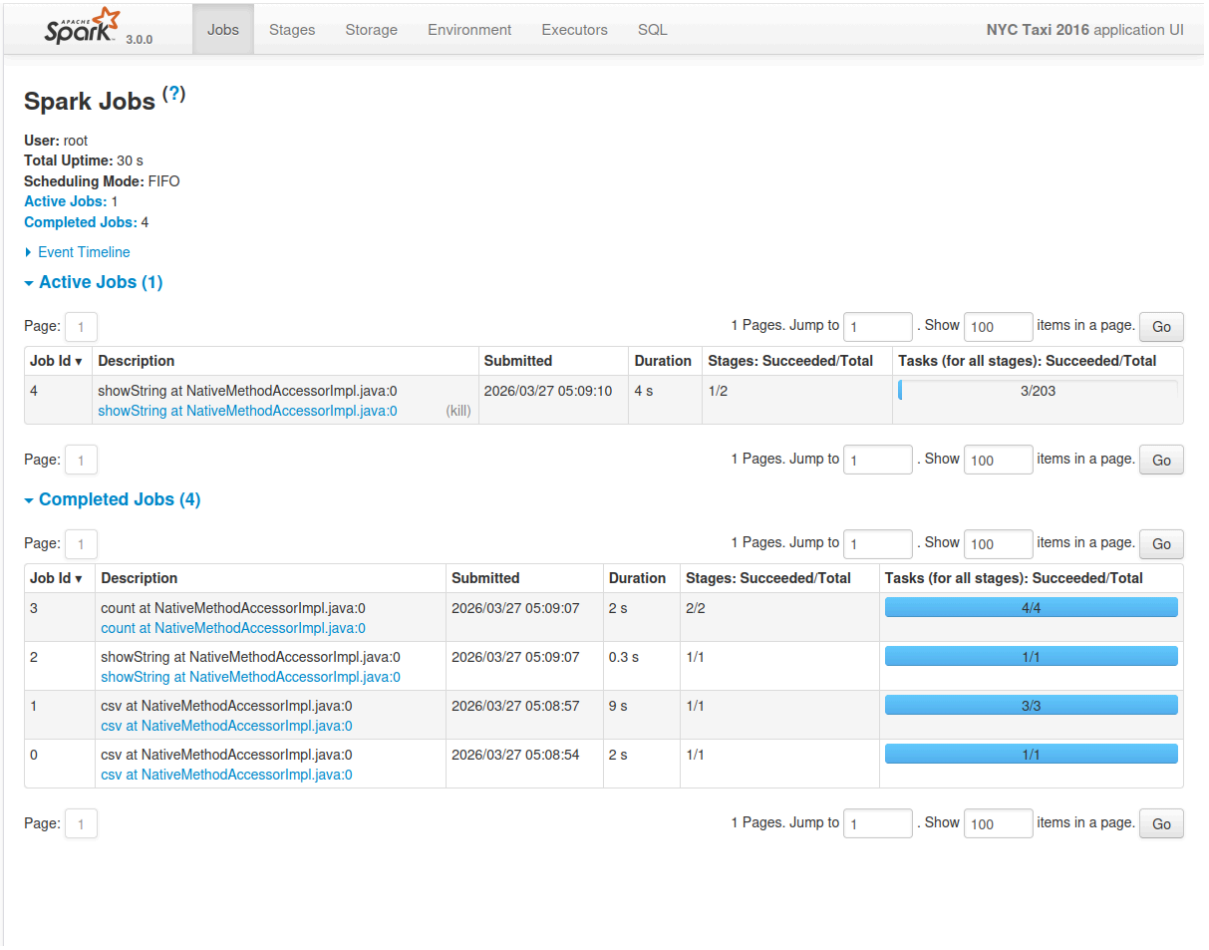
```
26/03/27 04:50:59 INFO ResourceUtils: ==============================================================
26/03/27 04:50:59 INFO SparkContext: Submitted application: NYC Taxi 2016
26/03/27 04:50:59 INFO SecurityManager: Changing view acls to: root
26/03/27 04:50:59 INFO SecurityManager: Changing modify acls to: root
26/03/27 04:50:59 INFO SecurityManager: Changing view acls groups to:
26/03/27 04:50:59 INFO SecurityManager: Changing modify acls groups to:
26/03/27 04:50:59 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users  with view permissions: Set(root); groups with view permissions: Set(); users  with modify pe
rmissions: Set(root); groups with modify permissions: Set()
26/03/27 04:50:59 INFO Utils: Successfully started service 'sparkDriver' on port 34817.
26/03/27 04:50:59 INFO SparkEnv: Registering MapOutputTracker
26/03/27 04:50:59 INFO SparkEnv: Registering BlockManagerMaster
26/03/27 04:50:59 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/03/27 04:50:59 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/03/27 04:50:59 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/03/27 04:50:59 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-fb61b24a-aa0e-4145-9b9e-33392593c714
26/03/27 04:50:59 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/03/27 04:50:59 INFO SparkEnv: Registering OutputCommitCoordinator
26/03/27 04:50:59 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/03/27 04:50:59 INFO SparkUI: Bound SparkUI to 0.0.0.0, and started at http://33507b646d2d:4040
26/03/27 04:51:00 INFO StandaloneAppClient$ClientEndpoint: Connecting to master spark://spark-master:7077...
26/03/27 04:51:00 INFO TransportClientFactory: Successfully created connection to spark-master/172.20.0.4:7077 after 52 ms (0 ms spent in bootstraps)
26/03/27 04:51:00 INFO StandaloneSchedulerBackend: Connected to Spark cluster with app ID app-20260327045100-0001
26/03/27 04:51:00 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260327045100-0001/0 on worker-20260327043814-172.20.0.8-44321 (172.20.0.8:44321) with 1 core(s)
26/03/27 04:51:00 INFO StandaloneSchedulerBackend: Granted executor ID app-20260327045100-0001/0 on hostPort 172.20.0.8:44321 with 1 core(s), 512.0 MiB RAM
26/03/27 04:51:00 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260327045100-0001/1 on worker-20260327043814-172.20.0.7-33437 (172.20.0.7:33437) with 1 core(s)
26/03/27 04:51:00 INFO StandaloneSchedulerBackend: Granted executor ID app-20260327045100-0001/1 on hostPort 172.20.0.7:33437 with 1 core(s), 512.0 MiB RAM
26/03/27 04:51:00 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260327045100-0001/2 on worker-20260327043813-172.20.0.6-39689 (172.20.0.6:39689) with 1 core(s)
26/03/27 04:51:00 INFO StandaloneSchedulerBackend: Granted executor ID app-20260327045100-0001/2 on hostPort 172.20.0.6:39689 with 1 core(s), 512.0 MiB RAM
26/03/27 04:51:00 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260327045100-0001/0 is now RUNNING
26/03/27 04:51:00 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260327045100-0001/1 is now RUNNING
26/03/27 04:51:00 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 42051.
26/03/27 04:51:00 INFO NettyBlockTransferService: Server created on 33507b646d2d:42051
26/03/27 04:51:00 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260327045100-0001/2 is now RUNNING
26/03/27 04:51:00 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/03/27 04:51:00 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, 33507b646d2d, 42051, None)
26/03/27 04:51:00 INFO BlockManagerMasterEndpoint: Registering block manager 33507b646d2d:42051 with 366.3 MiB RAM, BlockManagerId(driver, 33507b646d2d, 42051, None)
26/03/27 04:51:00 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 33507b646d2d, 42051, None)
26/03/27 04:51:00 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 33507b646d2d, 42051, None)
26/03/27 04:51:00 INFO StandaloneSchedulerBackend: SchedulerBackend is ready for scheduling beginning after reached minRegisteredResourcesRatio: 0.0
/spark/python/lib/pyspark.zip/pyspark/context.py:220: DeprecationWarning: Support for Python 2 and Python 3 prior to version 3.6 is deprecated as of Spark 3.0. See also the plan for dropping Python 2
 support at https://spark.apache.org/news/plan-for-dropping-python-2-support.html.
  DeprecationWarning)
26/03/27 04:51:01 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir ('file:/spark-warehouse').
26/03/27 04:51:01 INFO SharedState: Warehouse path is 'file:/spark-warehouse'.

===== SCHEMA =====
root
 |-- VendorID: integer (nullable = true)
 |-- tpep_pickup_datetime: string (nullable = true)
```

```
 |-- tpep_dropoff_datetime: string (nullable = true)
 |-- passenger_count: integer (nullable = true)
 |-- trip_distance: double (nullable = true)
 |-- pickup_longitude: double (nullable = true)
 |-- pickup_latitude: double (nullable = true)
 |-- RatecodeID: integer (nullable = true)
 |-- store_and_fwd_flag: string (nullable = true)
 |-- dropoff_longitude: double (nullable = true)
 |-- dropoff_latitude: double (nullable = true)
 |-- payment_type: integer (nullable = true)
 |-- fare_amount: double (nullable = true)
 |-- extra: double (nullable = true)
 |-- mta_tax: double (nullable = true)
 |-- tip_amount: double (nullable = true)
 |-- tolls_amount: double (nullable = true)
 |-- improvement_surcharge: double (nullable = true)
 |-- total_amount: double (nullable = true)

===== 5 DONG DAU =====
+--------+--------------------+---------------------+---------------+------------+-------------------+-------------------+----------+-------------------+-------------------+-------------------+----------
--+-----------+-----+-------+----------+-----------+--------------------+------------+
|VendorID|tpep_pickup_datetime|tpep_dropoff_datetime|passenger_count|trip_distance|pickup_longitude  |pickup_latitude   |RatecodeID|store_and_fwd_flag|dropoff_longitude |dropoff_latitude  |payment_ty
pe|fare_amount|extra|mta_tax|tip_amount|tolls_amount|improvement_surcharge|total_amount|
+--------+--------------------+---------------------+---------------+------------+-------------------+-------------------+----------+-------------------+-------------------+-------------------+----------
--+-----------+-----+-------+----------+-----------+--------------------+------------+
|1       |2016-03-01 00:00:04 |2016-03-01 00:03:30  |2              |0.5         |-73.9837875366211 |40.73189163208008 |1         |N                 |-73.97581481933594|40.728519439697266|1
  |4.5        |0.5  |0.5    |1.15       |0.0         |0.3                 |6.95        |
|2       |2016-03-01 00:00:13 |2016-03-01 00:04:56  |5              |0.79        |-73.99179077148438|40.74918746948242 |1         |N                 |-73.98950958251953|40.75777816772461 |1
  |5.5        |0.5  |0.5    |1.36       |0.0         |0.3                 |8.16        |
|2       |2016-03-01 00:00:20 |2016-03-01 00:11:56  |1              |1.84        |-73.99774932861328|40.75659942626953 |1         |N                 |-74.00470733642578|40.73453140258789 |1
  |8.5        |0.5  |0.5    |2.45       |0.0         |0.3                 |12.25       |
|2       |2016-03-01 00:00:27 |2016-03-01 00:17:52  |1              |10.2        |-73.8745346069336 |40.774112701416016|1         |N                 |-74.003662109375  |40.715911865234375|1
  |29.0       |0.5  |0.5    |6.06       |0.0         |0.3                 |36.36       |
|2       |2016-03-01 00:00:32 |2016-03-01 00:08:59  |1              |2.46        |-73.97491455078125|40.76174545288086 |1         |N                 |-73.98908996582031|40.732303619384766|1
  |9.5        |0.5  |0.5    |1.0        |0.0         |0.3                 |11.8        |
+--------+--------------------+---------------------+---------------+------------+-------------------+-------------------+----------+-------------------+-------------------+-------------------+----------
--+-----------+-----+-------+----------+-----------+--------------------+------------+
only showing top 5 rows

===== TONG SO DONG =====
610547
===== TRUNG BINH TIP THEO LOAI THANH TOAN =====
+------------+-------+---------+---------+
|payment_type|avg_tip|avg_total|so_chuyen|
+------------+-------+---------+---------+
|           1|   2.69|    17.43|   407014|
|           2|    0.0|    13.07|   200371|
|           3|  -0.01|    14.38|     2380|
|           4|    0.0|    13.13|      782|
+------------+-------+---------+---------+
```



Spark Master at spark://33507b646d2d:7077

URL: spark://33507b646d2d:7077
Alive Workers: 3
Cores in use: 3 Total, 3 Used
Memory in use: 3.0 GiB Total, 1536.0 MiB Used
Resources in use:
Applications: 1 Running, 8 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE

Workers (3)

Worker Id	Address	State	Cores	Memory	Resources
worker-20260327043813-172.20.0.6-39689	172.20.0.6:39689	ALIVE	1 (1 Used)	1024.0 MiB (512.0 MiB Used)	
worker-20260327043814-172.20.0.7-33437	172.20.0.7:33437	ALIVE	1 (1 Used)	1024.0 MiB (512.0 MiB Used)	
worker-20260327043814-172.20.0.8-44321	172.20.0.8:44321	ALIVE	1 (1 Used)	1024.0 MiB (512.0 MiB Used)	

Running Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260327050845-0008 (kill)	NYC Taxi 2016	3	512.0 MiB		2026/03/27 05:08:45	root	RUNNING	35 s

Completed Applications (8)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260327050638-0007	WordCount	3	512.0 MiB		2026/03/27 05:06:38	root	FINISHED	11 s
app-20260327050355-0006	WordCount	3	512.0 MiB		2026/03/27 05:03:55	root	FINISHED	6 s
app-20260327050234-0005	WordCount	3	512.0 MiB		2026/03/27 05:02:34	root	FINISHED	7 s
app-20260327050120-0004	WordCount	3	512.0 MiB		2026/03/27 05:01:20	root	FINISHED	7 s
app-20260327050002-0003	WordCount	3	512.0 MiB		2026/03/27 05:00:02	root	FINISHED	7 s
app-20260327045510-0002	WordCount	3	512.0 MiB		2026/03/27 04:55:10	root	FINISHED	8 s
app-20260327045100-0001	NYC Taxi 2016	3	512.0 MiB		2026/03/27 04:51:00	root	FINISHED	33 s
app-20260327044723-0000	WordCount	3	512.0 MiB		2026/03/27 04:47:23	root	FINISHED	14 s